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Table 4: Expert-labeled examples from annotation study

Example 1

Idea: Develop a **natural language processing classifier designed to improve scientific paper revisions** by automatically identifying and categorizing reviewer comments that are most likely to lead to substantial and actionable revisions. The system would be trained on a **manually-labeled dataset analysis** of scientific review comments and the corresponding paper edits, leveraging features such as linguistic cues, sentiment, and comment specificity to predict the likelihood of a comment being acted upon. This classifier could then be used to prioritize reviewer feedback, helping authors focus on the most impactful suggestions first.

Most Relevant Papers:

1. [ARIES: A Corpus of Scientific Paper Edits Made in Response to Peer Reviews](#)
2. [Can large language models provide useful feedback on research papers?](#)
3. [A Dataset of Peer Reviews \(PeerRead\): Collection, Insights and NLP Applications](#)
4. [arXivEdits: Understanding the Human Revision Process in Scientific Writing](#)
5. [Characterizing Text Revisions to Better Support Collaborative](#)
6. [Can We Automate Scientific Reviewing?](#)
7. [DeepReviewer: Collaborative Grammar & Innovation Neural Network for Paper Review](#)
8. [Aspect-based Sentiment Analysis of Scientific Reviews](#)
9. [Aspect-based sentiment analysis of online peer reviews and prediction of paper acceptance](#)
10. [ReviVal: Towards Automatically Evaluating the Informativeness of Peer Reviews](#)

Reasoning: The idea is **novel** because it uniquely focuses on prioritizing reviewer comments for actionable revisions, which is not explicitly addressed in ARIES[1] or other works like ReviVal[10].

Example 2

Idea: Develop a **systematic review-based framework designed to align LLM evaluation with human preferences**, ensuring that evaluation criteria are continuously refined based on comprehensive reviews of user feedback and emerging model behaviors. This framework will utilize **content analysis of user interactions and feedback** to identify patterns and areas of improvement. The effectiveness of this framework will be assessed through a **qualitative study** involving iterative cycles of user feedback and criteria refinement.

Most Relevant Papers:

1. [EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined Criteria](#)
2. [Humanely: Human evaluation of LLM yield, using a novel web-based evaluation tool](#)
3. [Evaluation of Code Generation for Simulating Participant Behavior in Experience Sampling Method by Iterative In-Context Learning of a Large Language Model](#)
4. [Human-Centered Evaluation and Auditing of Language Models](#)
5. [Aligning Model Evaluations with Human Preferences: Mitigating Token Count Bias in Language Model Assessments](#)
6. [Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences](#)
7. [Human-Centered Design Recommendations for LLM-as-a-judge](#)
8. [CheckEval: Robust Evaluation Framework using Large Language Model via Checklist](#)
9. [Discovering Language Model Behaviors with Model-Written Evaluations](#)
10. [Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models](#)

Reasoning: The idea is **not novel** because it closely resembles existing frameworks like EvalLM[1] and HumanELY[2], which already align LLM evaluations with human preferences using user-defined criteria and human feedback.